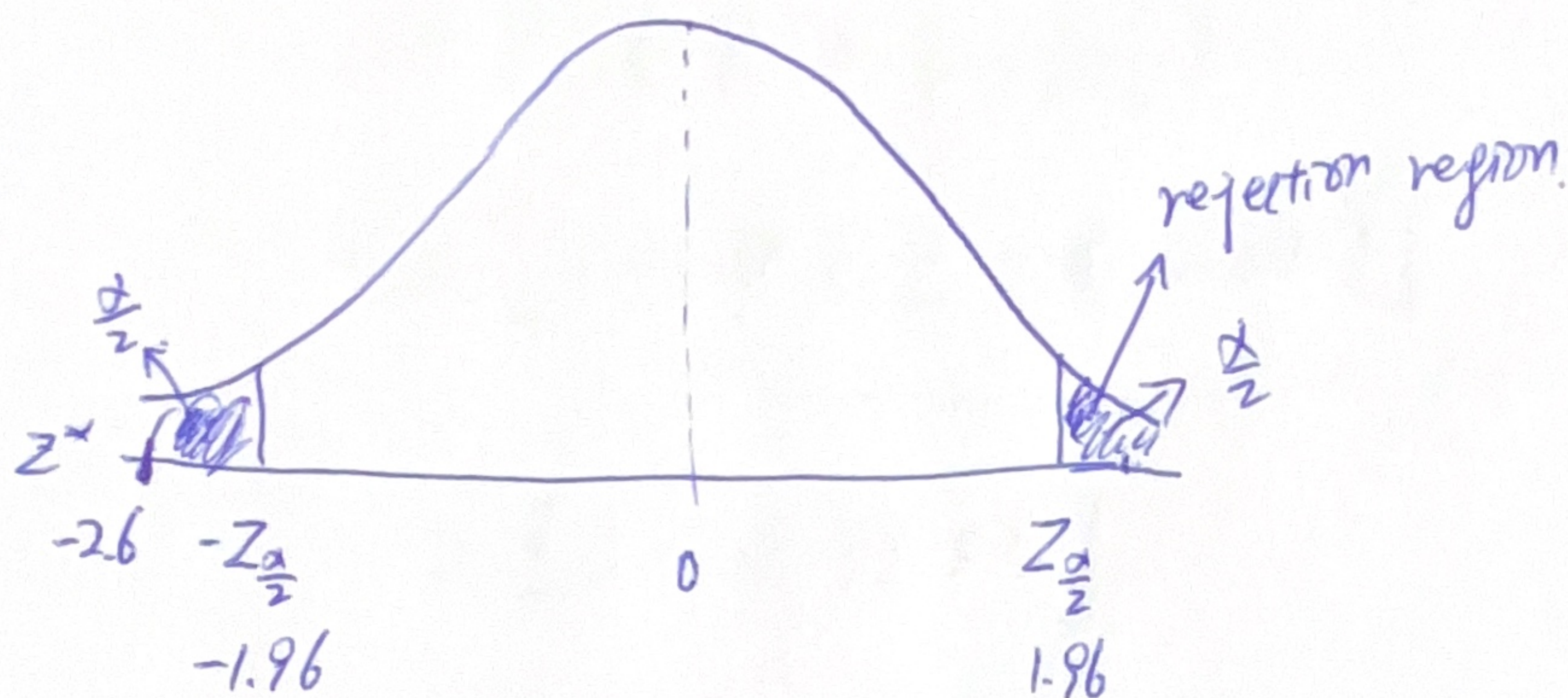


* How to understand the process?

④



P-value

Def'n: P-value = $P(\text{get a test statistic at least as extreme as you just observed} \mid H_0 \text{ is true})$

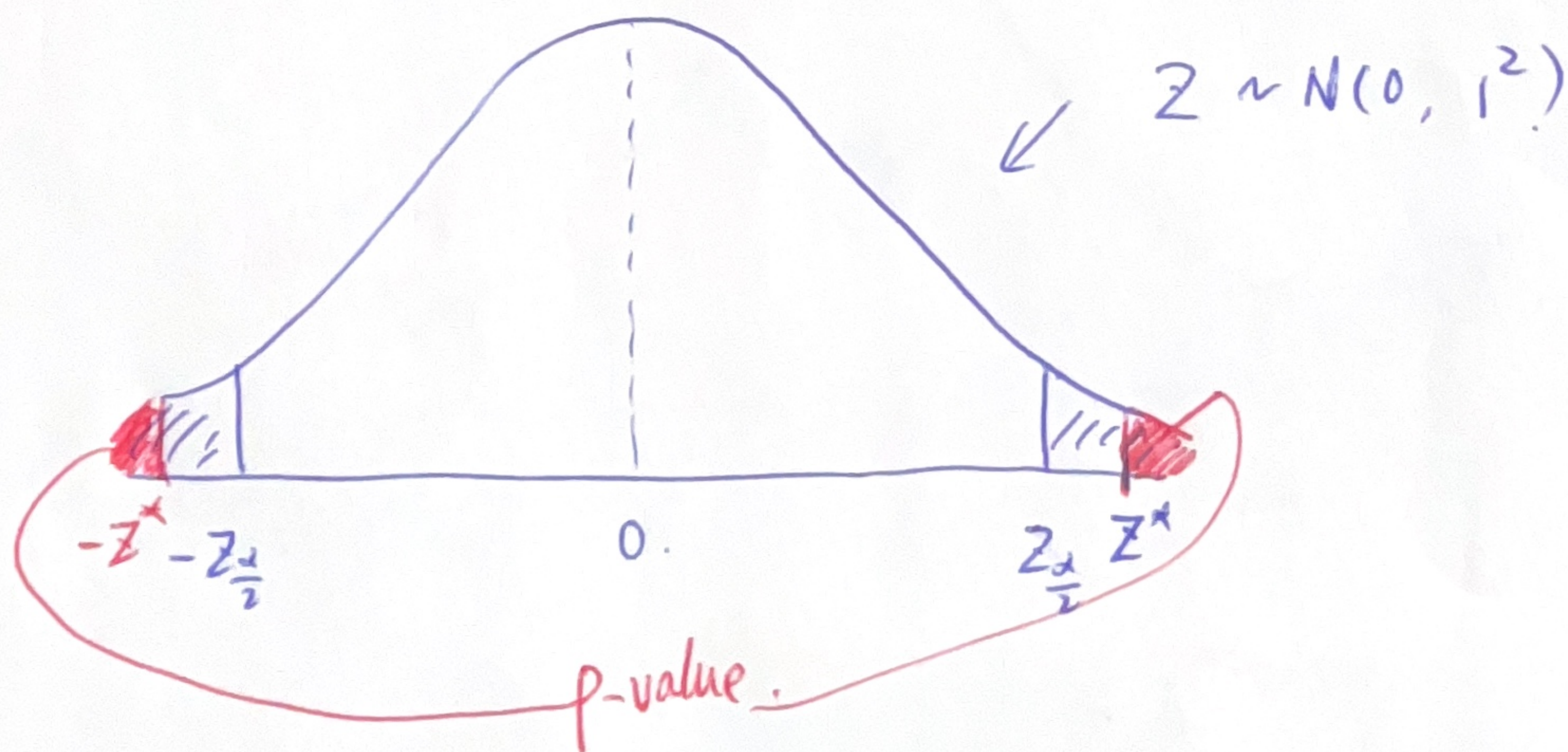
two-sided: $p\text{-value} = 2P(Z > |Z^*|)$

right-sided: $p\text{-value} = P(Z > Z^*)$

left-sided: $p\text{-value} = P(Z < Z^*)$

Rule: If $p\text{-value} < \alpha$, reject H_0

If $p\text{-value} \geq \alpha$, fail to reject H_0 .



$$b> \text{P-value} = 2 P(Z > |Z^*|)$$

$$= 2 P(Z > |-2.60|)$$

$$= 2 [1 - P(Z \leq 2.60)]$$

$$= 2 [1 - 0.9953]$$

$$= 0.0094$$

$$\text{P-value } 0.0094 < 0.05 = \alpha$$

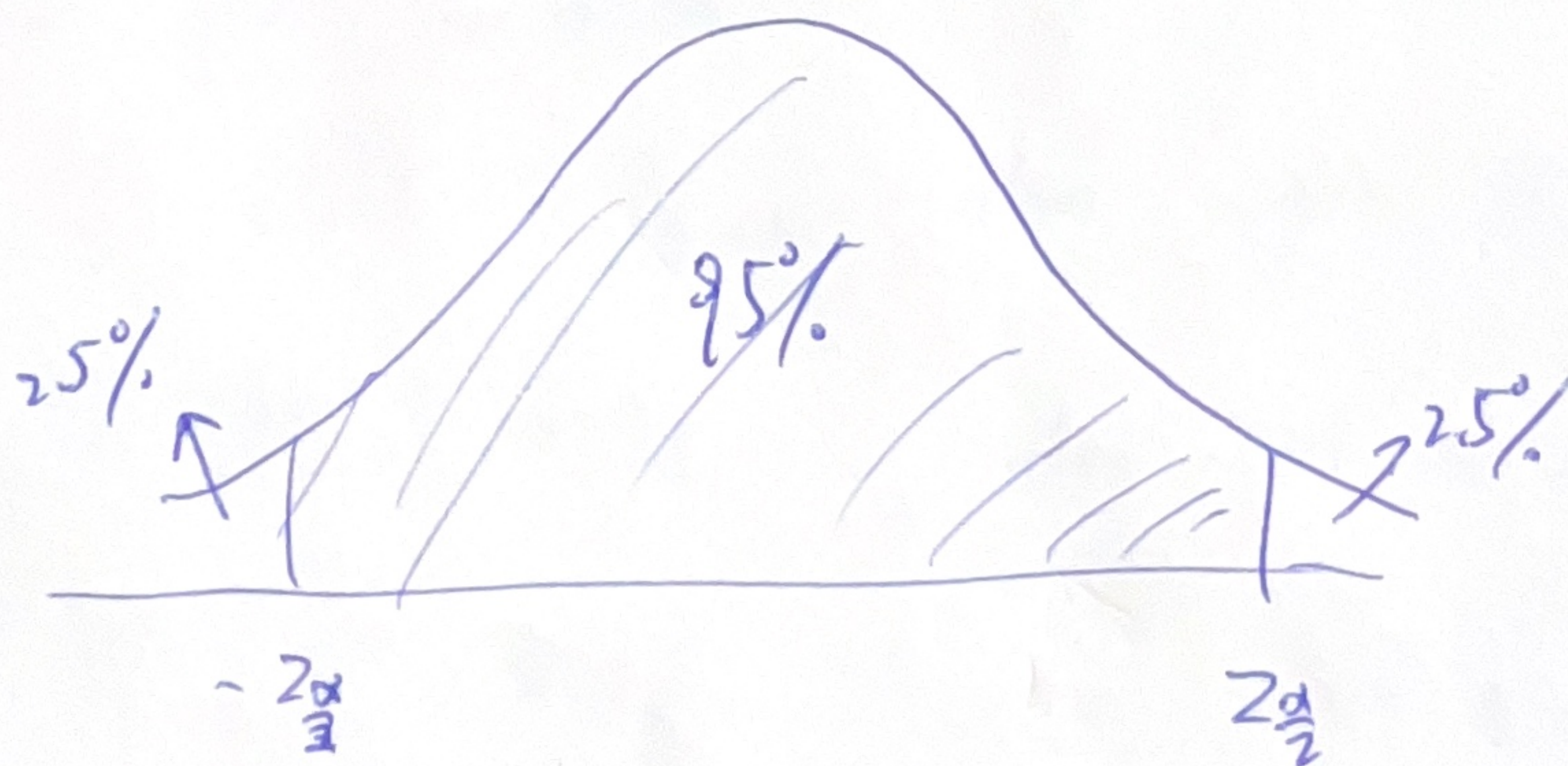
Therefore, reject $H_0: \mu = 16 \text{ oz}$.

We have strong enough evidence to support $H_a: \mu \neq 16 \text{ oz}$.

c> A 95% CI for μ .

$$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} = 15.974 \pm 1.96 \frac{0.05}{\sqrt{25}} = (15.9544, 15.9936)$$

16 is not in CI. the true mean μ is significantly different from 16 oz.



②

Note: If population / distribution is not normally distributed, but $n \geq 30$ then we still can do the (approximate) 3-step test, p-value and CI just like previous practices.

Hypothesis test for μ (σ unknown)

3-step. t test.

$$H_0: \mu = \mu_0 \quad \text{vs.} \quad H_a: \begin{cases} \mu \neq \mu_0 & \text{two-sided.} \\ \mu > \mu_0 & \text{right-sided} \\ \mu < \mu_0 & \text{left-sided.} \end{cases}$$

1> get the test statistic.

$$t^* = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

2> critical value.

$$\text{two-sided: } t_{\frac{\alpha}{2}, n-1}$$

$$\text{right-sided: } t_{\alpha, n-1}$$

$$\text{left-sided: } -t_{\alpha, n-1}$$

3> Decision and final conclusion

$$\text{two-sided: If } |t^*| > t_{\frac{\alpha}{2}, n-1}, \text{ then reject } H_0.$$

$$\text{If } |t^*| \leq t_{\frac{\alpha}{2}, n-1}, \text{ then fail to reject } H_0.$$

$$\text{right-sided: If } t^* > t_{\alpha, n-1} \text{ then reject } H_0.$$

$$\text{left-sided: If } t^* < -t_{\alpha, n-1}, \text{ then reject } H_0.$$

P-value for t-test.

③

two-sided: $p\text{-value} = 2 P(t_{n-1} > |t^*|)$

right-sided: $p\text{-value} = P(t_{n-1} > t^*)$

left-sided: $p\text{-value} = P(t_{n-1} < t^*)$

Rules: If $p\text{-value} < \alpha$, then reject H_0

If $p\text{-value} \geq \alpha$, then fail to reject H_0 .

Example: Amount of cheese on frozen pizzas has a normal distribution.
We R.S. 4 pizzas and found 5.4, 5.8, 5.5, 5.7 oz of cheese.

a) 3-step t-test for:

$$H_0: \mu = \underbrace{5.5}_{\mu_0} \quad \text{vs.} \quad H_a: \mu \neq 5.5.$$

$$\bar{x} = \frac{5.4 + 5.8 + 5.5 + 5.7}{4} = 5.6$$

$$s = \sqrt{\frac{1}{3} [(5.4 - 5.6)^2 + (5.8 - 5.6)^2 + (5.5 - 5.6)^2 + (5.7 - 5.6)^2]} \\ = 0.1826$$

$$n = 4.$$

▷ statistic

$$t^* = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{5.6 - 5.5}{0.1826/\sqrt{4}} = 1.095.$$

2) critical value. $\alpha = 0.05$

$$t_{\frac{\alpha}{2}, n-1} = t_{0.025, 3} = 3.182.$$

3) Decision and final conclusion.

$$|t^*| = |1.095| = 1.095 < 3.182 = t_{\frac{\alpha}{2}, n-1}.$$

Therefore, fail to reject H_0 .

We don't have strong enough evidence to support $H_a = \mu \neq 5.5$.

b) ~~4~~ p-value = $2P(t_{n-1} > |t^*|)$

$$= 2P(t_3 > 1.095) = 2 \times (0.15, 0.2)$$

$$\in (0.3, 0.4)$$

$$p\text{-value} > \alpha = 0.05$$

Therefore, fail to reject H_0 .

c) 95% CI for μ .

$$\bar{x} \pm t_{n-1, \frac{\alpha}{2}} \frac{s}{\sqrt{n}} = 5.6 \pm \frac{t_{0.025, 3}}{3.182} \times \frac{0.1826}{\sqrt{4}} = (5.31, 5.89)$$

d) Test $H_0: \mu = 5.3$

vs.

$H_a: \mu > 5.3$

right-sided test.

3-step t-test

1) test statistic.

$$t^* = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{5.6 - 5.3}{0.1826/\sqrt{4}} = 3.286$$